

Similarity-Distance-Magnitude Activations

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Reexpress AI

Findings of the Association for Computational Linguistics: ACL 2026

8 minute overview

Setting

Task: Multi-class Classification with language models (LMs)

Classify the input $\mathbf{x} \in \mathcal{X}$ into one of C discrete classes,
 $y \in \mathcal{Y} = \{1, \dots, C\}$

Data: Labeled training set, $\mathcal{D}_{\text{tr}} = \{(\mathbf{x}_n, y_n)\}_{n=1}^N$ and a corresponding held-out calibration set, $\mathcal{D}_{\text{ca}}$, drawn from the same distribution.

Setting: The parameters of the underlying LM are accessible, but held fixed. A final-layer adaptor has access to the frozen hidden states across all sequence positions.

Seek: Prediction, $\hat{y}$, and an estimate of $p(y \mid \mathbf{x})$, the predictive uncertainty of that classification decision.

Calibration: Class- and prediction-conditional accuracy at least α , a value near 1 (e.g., $\alpha = 0.95$)

Necessity of Actionable Interpretability for Language Models

Decision-making with language models (LMs).

High accuracy for point predictions, $\hat{y}$, is necessary but not sufficient

➡ Also, need an estimate of $p(y | \mathbf{x})$, the predictive uncertainty

Challenge.

The ESTIMATOR of the predictive uncertainty cannot itself be a black-box

- Parameters of neural networks are *non-identifiable*
- Softmax over output logits is a highly lossy compression of the *epistemic* (reducible) uncertainty
- Calibration is largely independent of model scale

Stakes are maximally high given the ubiquity of LMs in decision-making

Actionable Interpretability for Language Models

Approach.

Shift locus of interpretability: *From* PARAMETERS *to* ESTIMATORS

A Memory Layer for Actionable Interpretability.

SIMILARITY-DISTANCE-MAGNITUDE (SDM) ESTIMATORS provide

- ① $\hat{p}(y \mid \mathbf{x})$, a robust estimate of the predictive uncertainty
- ② Interpretability-by-exemplar against a database of labeled examples (*introspection*)
- ③ Local *updatability* via changes to $\mathcal{D}_{\text{tr}}$ and $\mathcal{D}_{\text{ca}}$.

Existing formative work

2016-2025.

A selection of key existing methods for explainable and interpretable AI

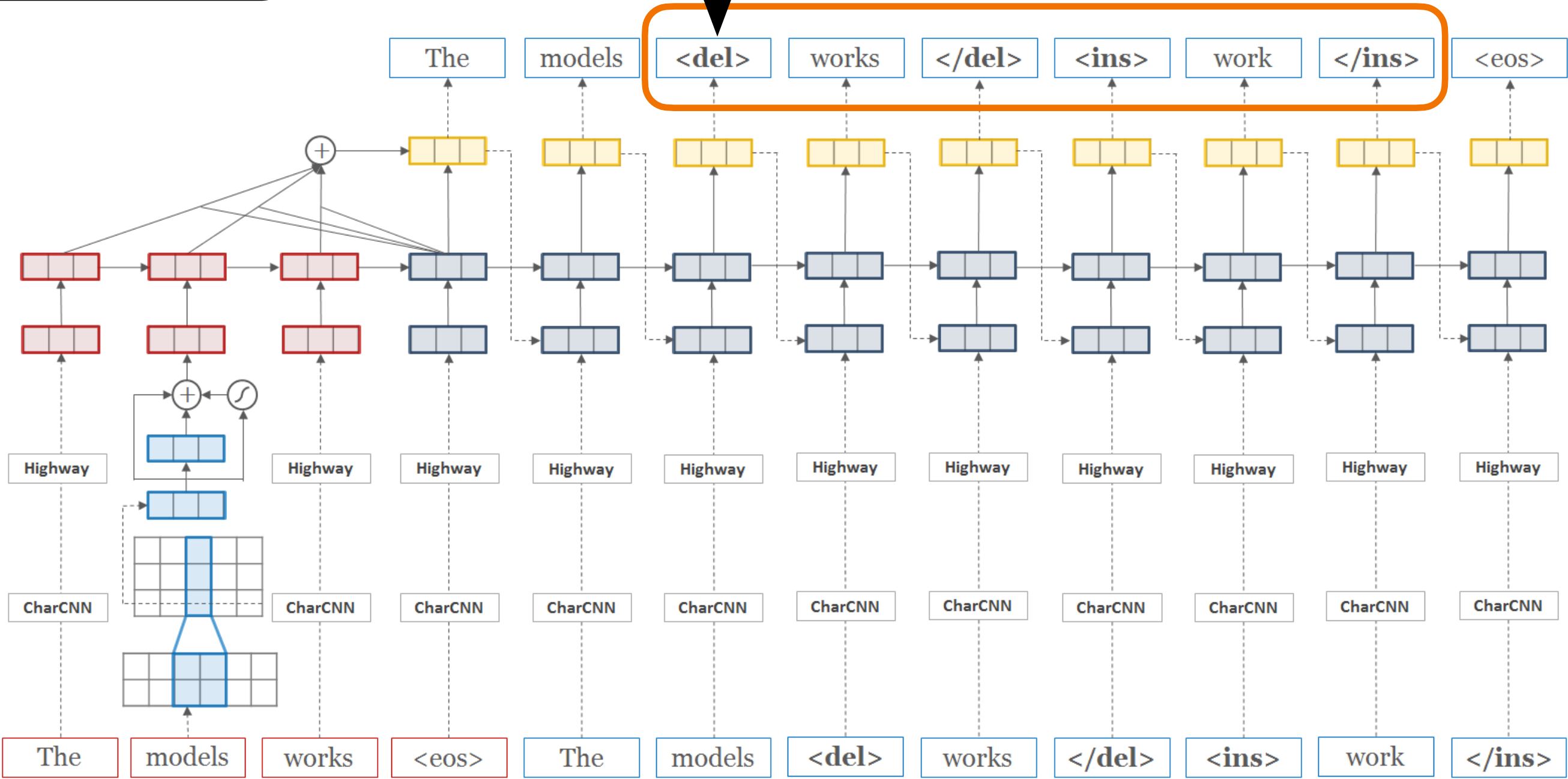
2016: Endogenous modeling of LM explanations

Self-explanations/self-rationalizations.

Model explainable classification as supervised sequence generation.

Elements of actionable interpretability.

The weights of the control tokens can be post-hoc adjusted at test-time to guide explanation generation.

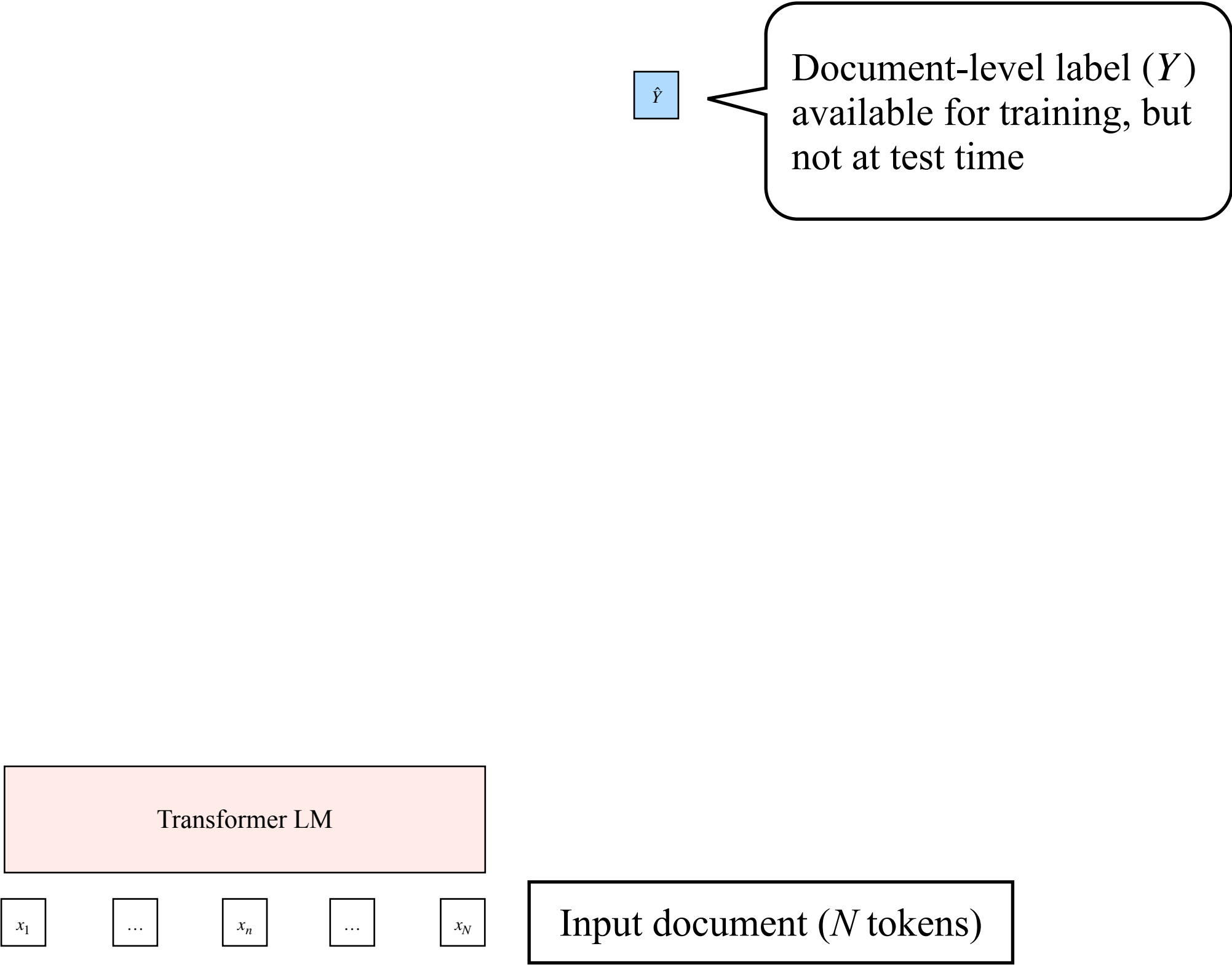


Limitations.

Absence of robust calibration makes the model brittle to distribution shifts.
 Not clear how the training instances relate to new, unseen test examples.

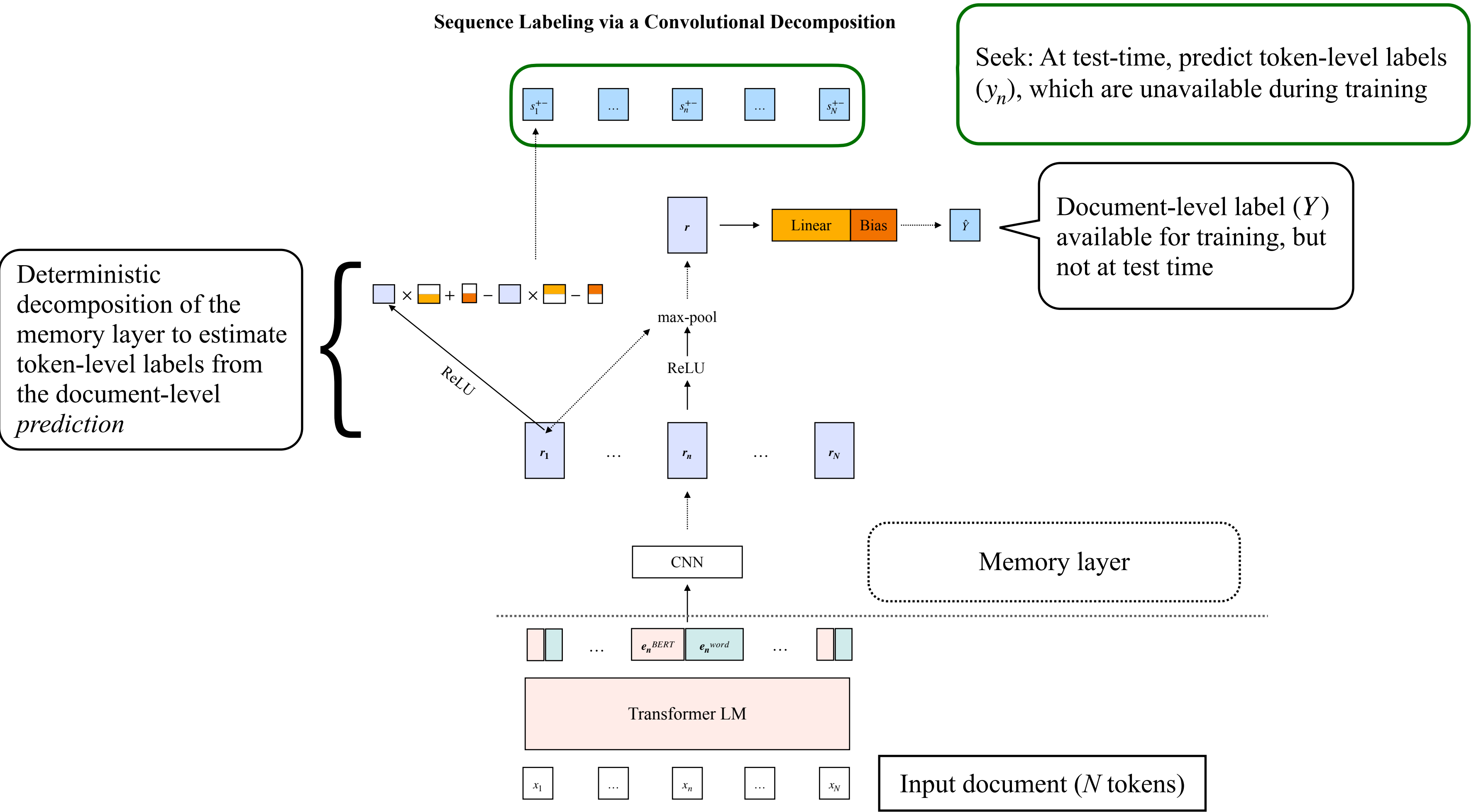
Schmaltz, Kim, Rush, and Shieber. 2016. Sentence-Level Grammatical Error Identification as Sequence-to-Sequence Correction.

2021: LMs as decomposable, hidden instance-based metric learners



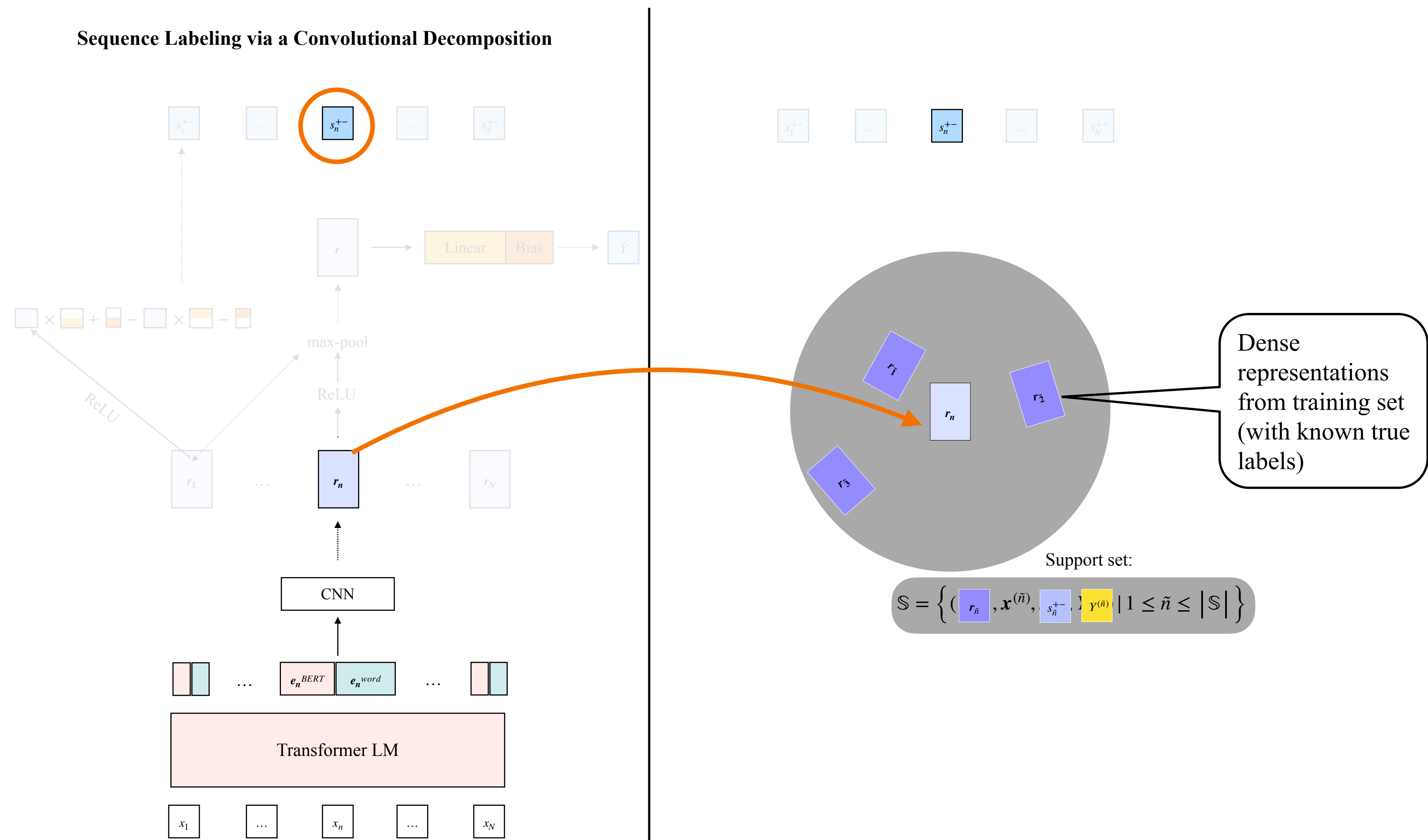
Schmaltz. 2021. Detecting Local Insights from Global Labels: Supervised & Zero-Shot Sequence Labeling via a Convolutional Decomposition.

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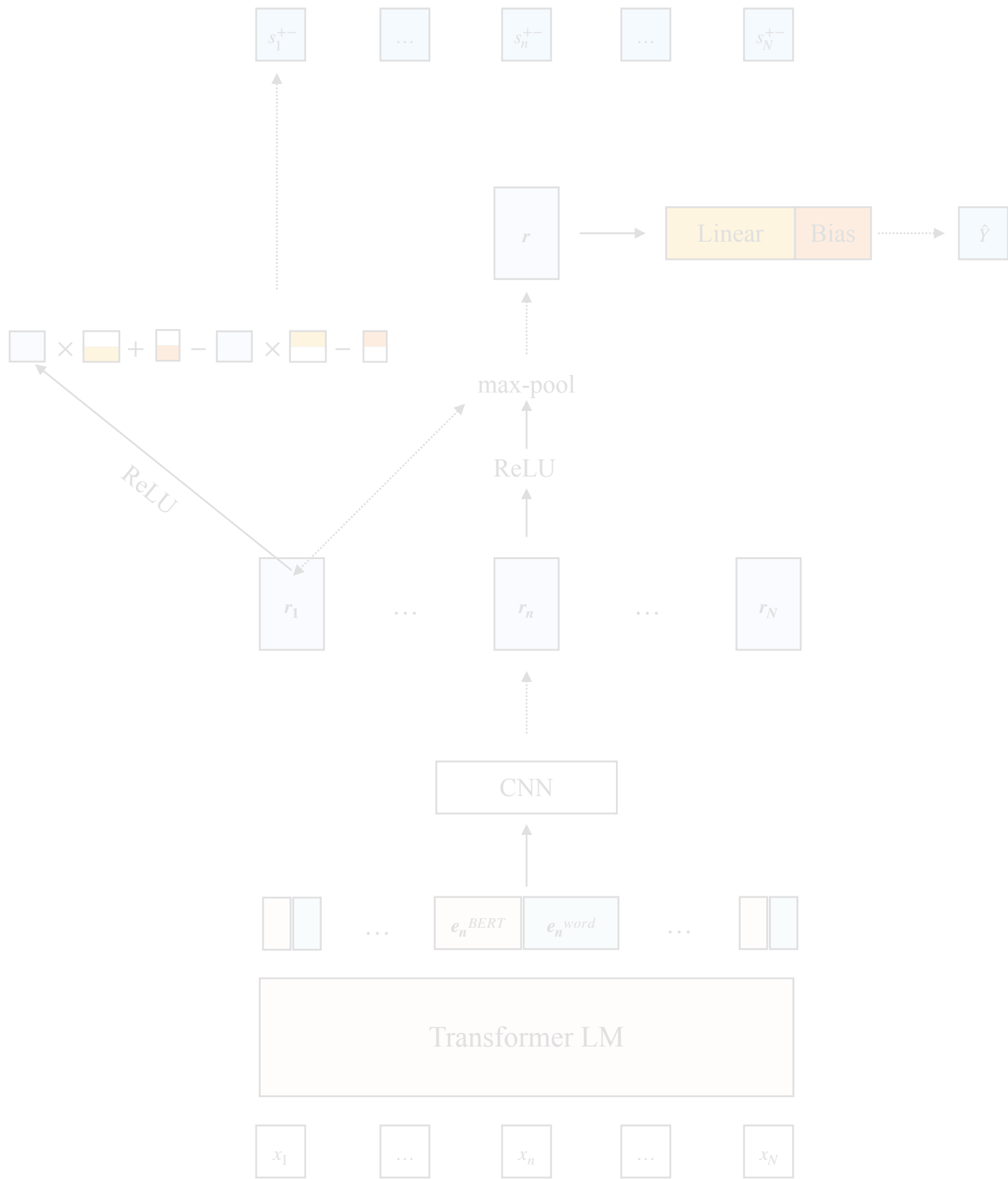
2021: LMs as decomposable, hidden instance-based metric learners



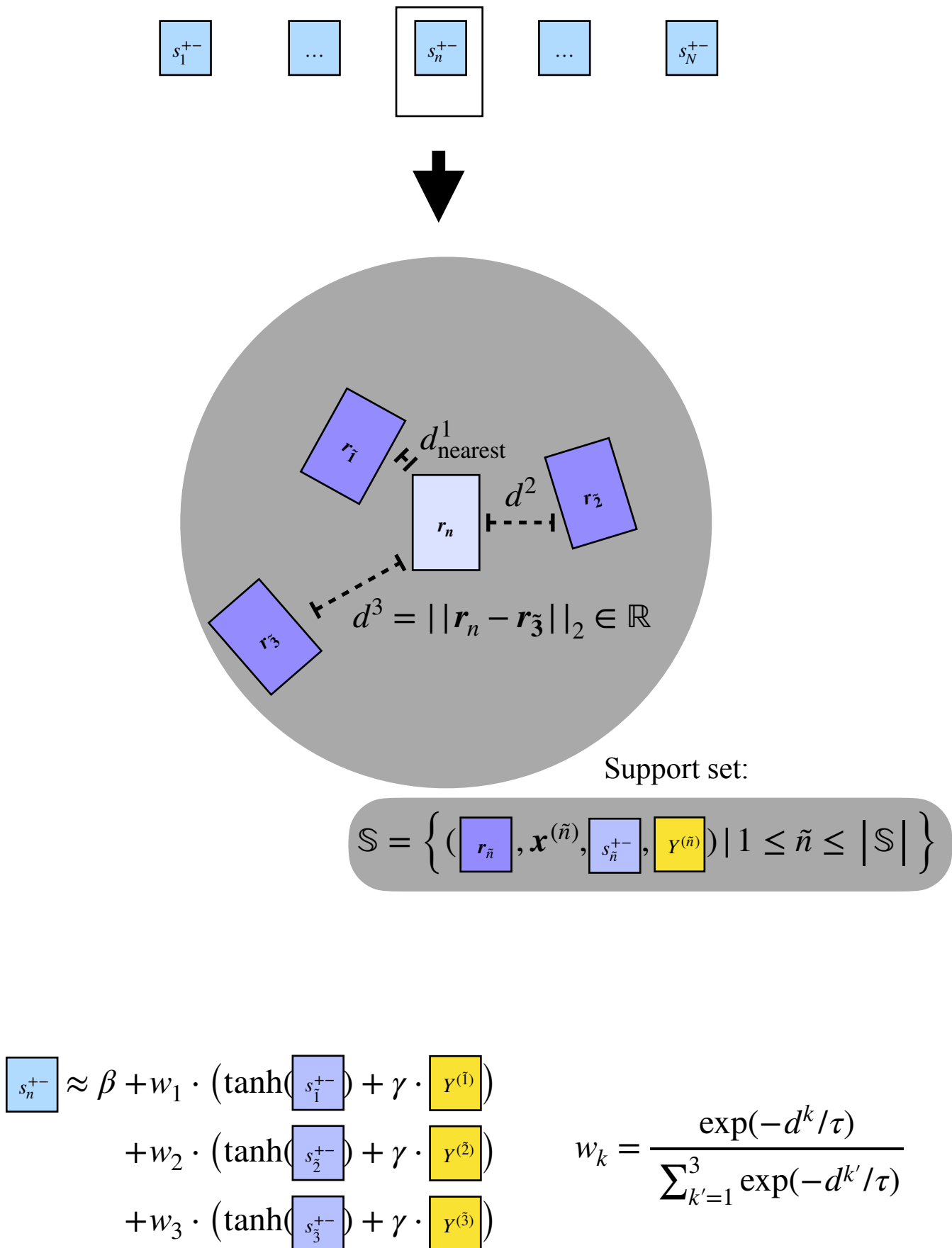
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Sequence Labeling via a Convolutional Decomposition



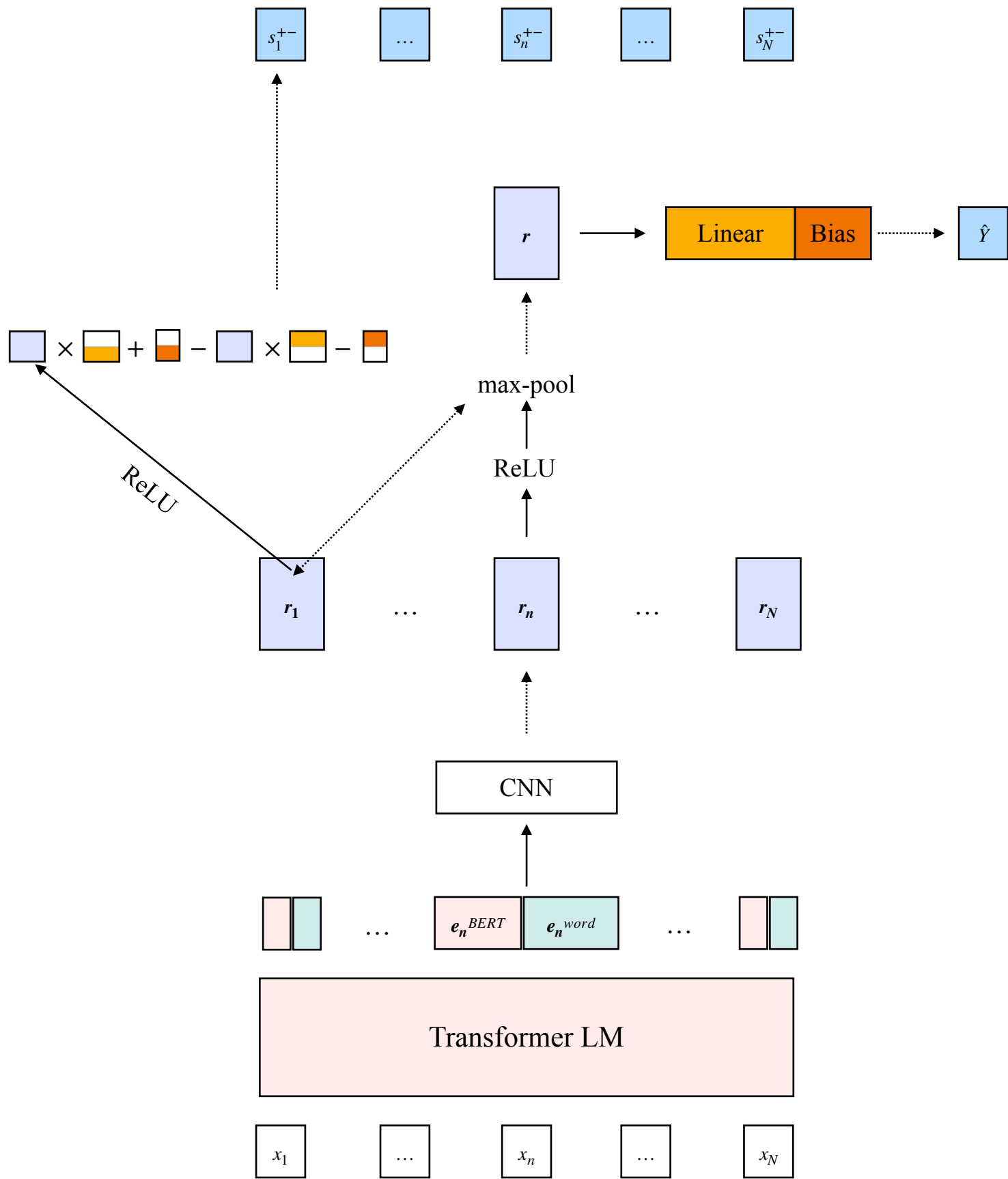
K-NN Approximation



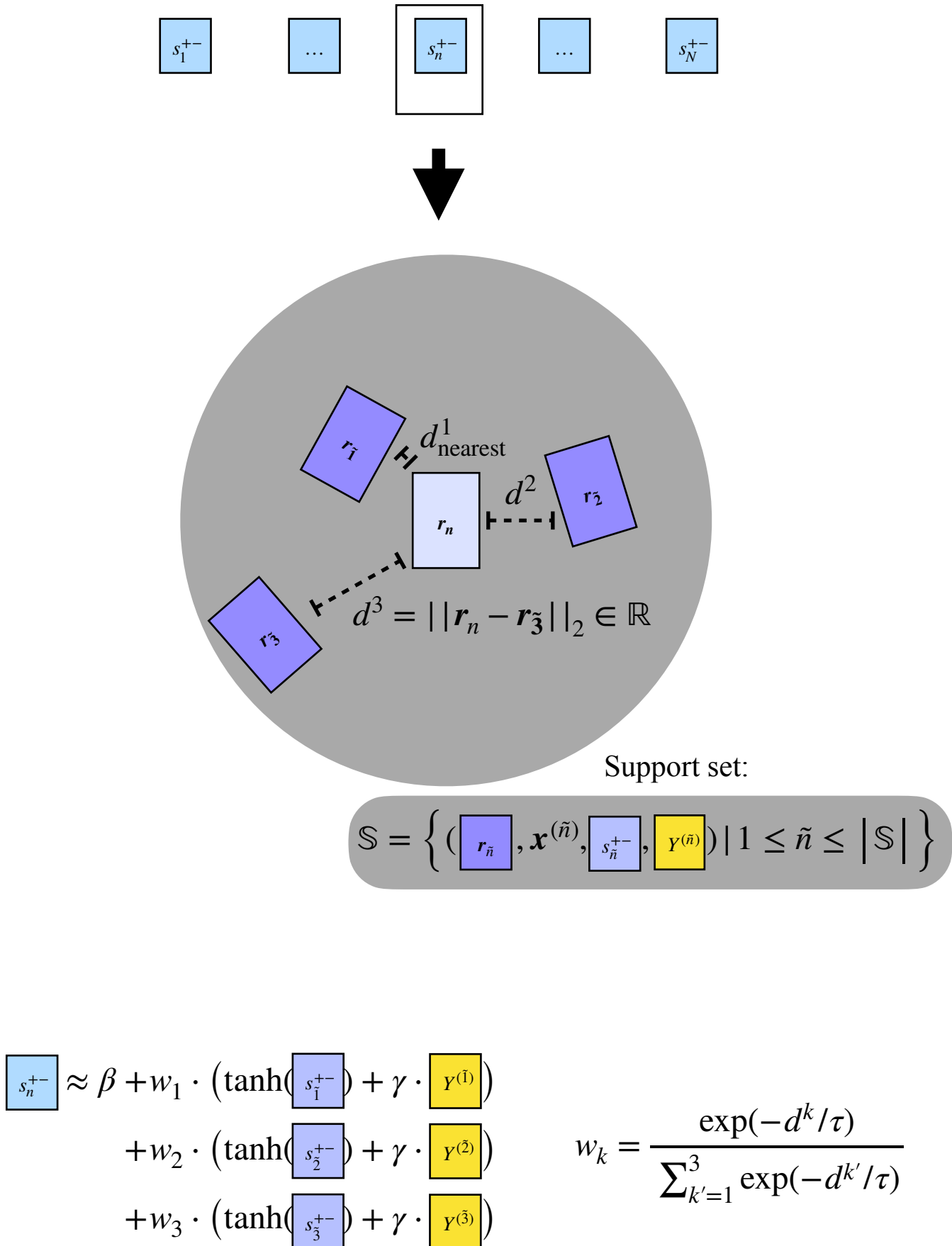
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Sequence Labeling via a Convolutional Decomposition



K-NN Approximation



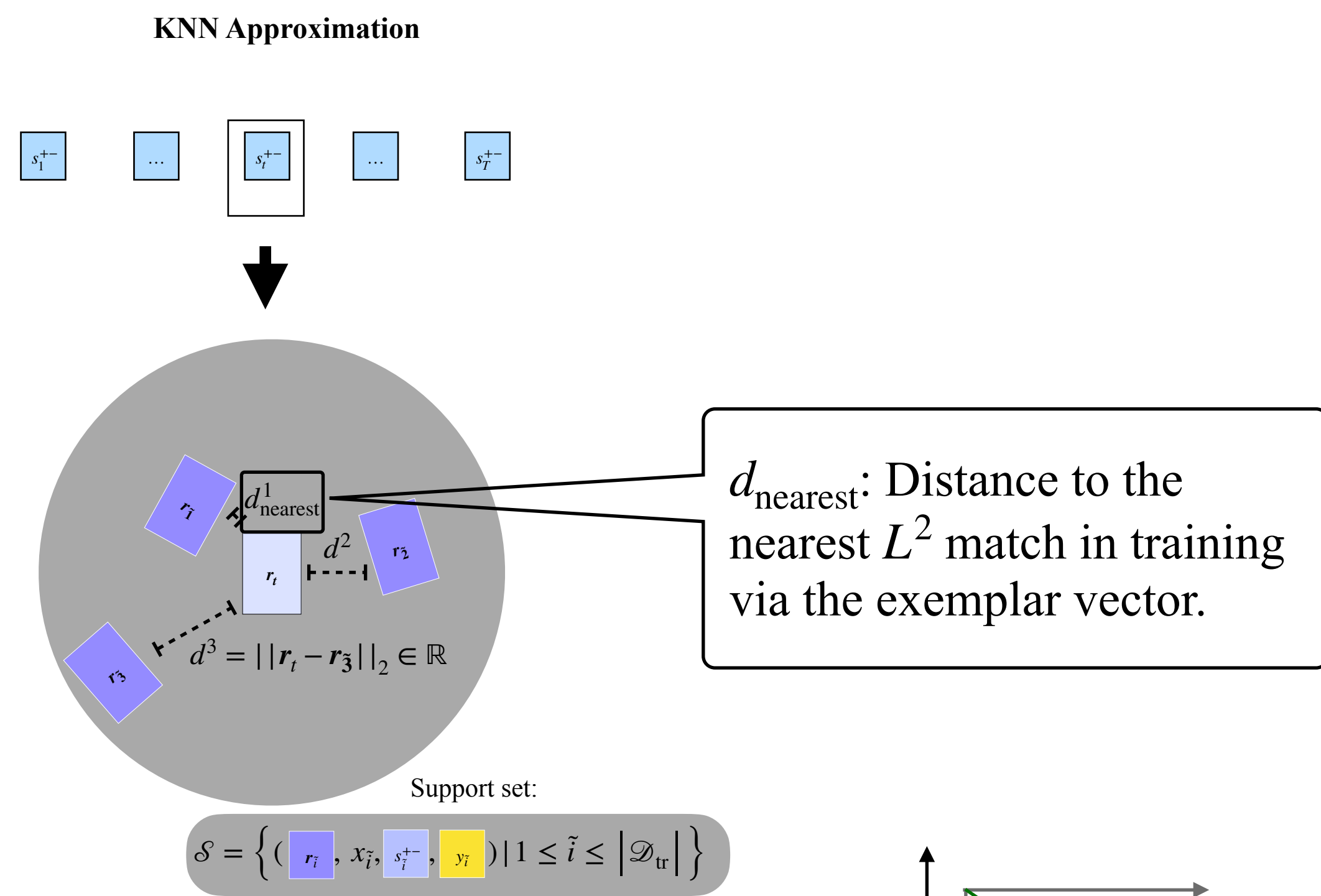
Schmaltz. 2021. Detecting Local Insights from Global Labels: Supervised & Zero-Shot Sequence Labeling via a Convolutional Decomposition.

Faithful approximations.

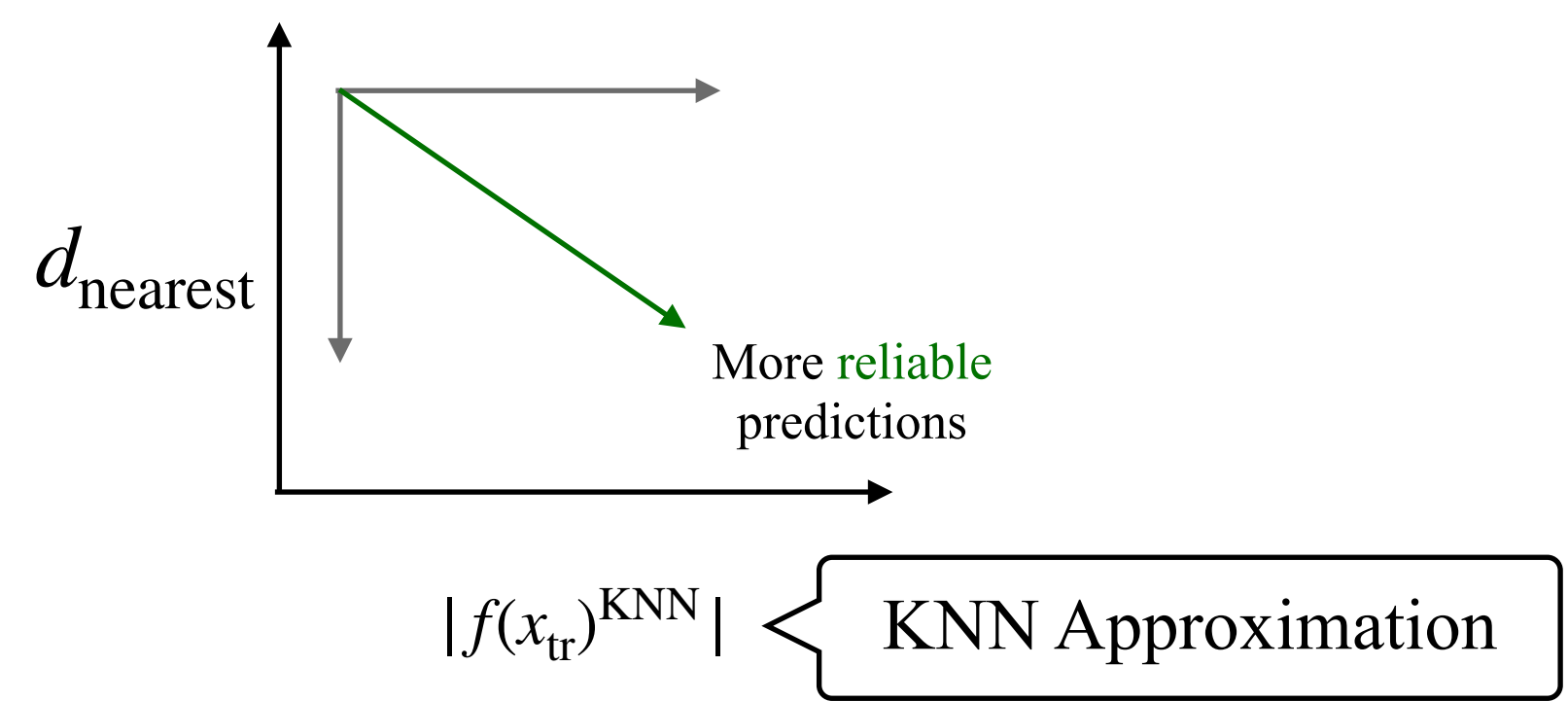
- The network can be closely approximated as a metric learner.
- Approximation errors follow an actionable pattern: They tend to be from regions of the distribution that are difficult for the original model to predict.
- Establishes an empirical basis for consistency approaches to uncertainty quantification for LMs.
- Provides instance-wise data valuation.
- Enables local updatability via changes to the support set without changing the LM’s weights.

Limitations.

- Independent moving parts: Multiple models to learn (memory layer + approximation)
- d_{nearest} is, in effect, exogenous to the approximation (since SOFTMAX is invariant under translation)
- Ad-hoc calibration



$$s_t^{+-} \approx \beta + w_1 \cdot \left(\tanh(s_1^{+-}) + \gamma \cdot y_1 \right) + w_2 \cdot \left(\tanh(s_2^{+-}) + \gamma \cdot y_2 \right) + w_3 \cdot \left(\tanh(s_3^{+-}) + \gamma \cdot y_3 \right)$$
$$w_k = \frac{\exp(-d^k/\eta)}{\sum_{k'=1}^3 \exp(-d^{k'}/\eta)}$$



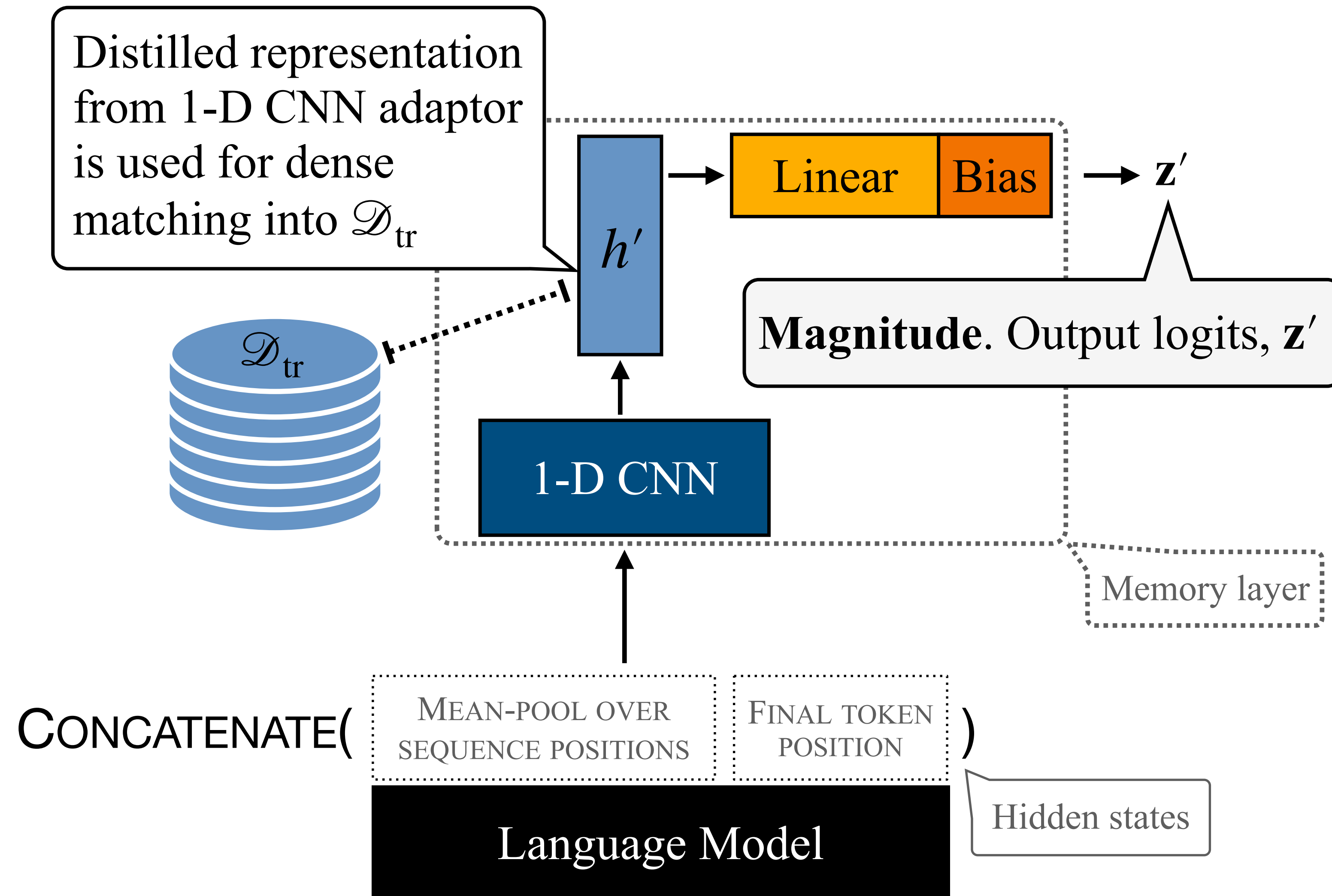
Origin of the phrase “actionable interpretability”

Schmaltz. 2021. Detecting Local Insights from Global Labels: Supervised & Zero-Shot Sequence Labeling via a Convolutional Decomposition.

Similarity-Distance-Magnitude (SDM) Activations

A Memory Layer for Actionable Interpretability.

SDM Activation: Memory Layer



SDM Activation replaces the final-layer SOFTMAX

$$\begin{aligned}\text{SDM}(\mathbf{z}')_i &= \frac{\text{Similarity}^{\text{Distance} \cdot \text{Magnitude}_i}}{\sum_{c=1}^C \text{Similarity}^{\text{Distance} \cdot \text{Magnitude}_c}} \\ &= \frac{(2 + q)^{d \cdot z'_i}}{\sum_{c=1}^C (2 + q)^{d \cdot z'_c}}, \quad 1 \leq i \leq C\end{aligned}$$

SDM Activation replaces the final-layer SOFTMAX

$$\begin{aligned} \text{SDM}(\mathbf{z}')_i &= \frac{\text{Similarity}^{\text{Distance} \cdot \text{Magnitude}_i}}{\sum_{c=1}^C \text{Similarity}^{\text{Distance} \cdot \text{Magnitude}_c}} \\ &= \frac{(2+q)^{d \cdot z'_i}}{\sum_{c=1}^C (2+q)^{d \cdot z'_c}}, \quad 1 \leq i \leq C \end{aligned}$$

Distance quantile, d .
The min controls for
class-wise skew in the
distribution of outliers.

$$d = \min \left(1 - \text{eCDF}_{\text{ca}}^{y=0}(d_{\text{Nearest}}), \right. \\ \left. 1 - \text{eCDF}_{\text{ca}}^{y=1}(d_{\text{Nearest}}) \right)$$

Test point: True
label, y , unknown

Test:
 h'

L^2 distance to nearest
training instance

d_{nearest}

Comparisons to $\mathcal{D}_{\text{tr}}$

$$[\hat{y} = \hat{y}_{(1)}^{\text{tr}}] \wedge [\hat{y}_{(1)}^{\text{tr}} = y_{(1)}^{\text{tr}}]$$

+

$$[\hat{y} = \hat{y}_{(2)}^{\text{tr}}] \wedge [\hat{y}_{(2)}^{\text{tr}} = y_{(2)}^{\text{tr}}]$$

$$[\hat{y} \neq \hat{y}_{(3)}^{\text{tr}}] \vee [\hat{y}_{(3)}^{\text{tr}} \neq y_{(3)}^{\text{tr}}]$$

Train:
1st match

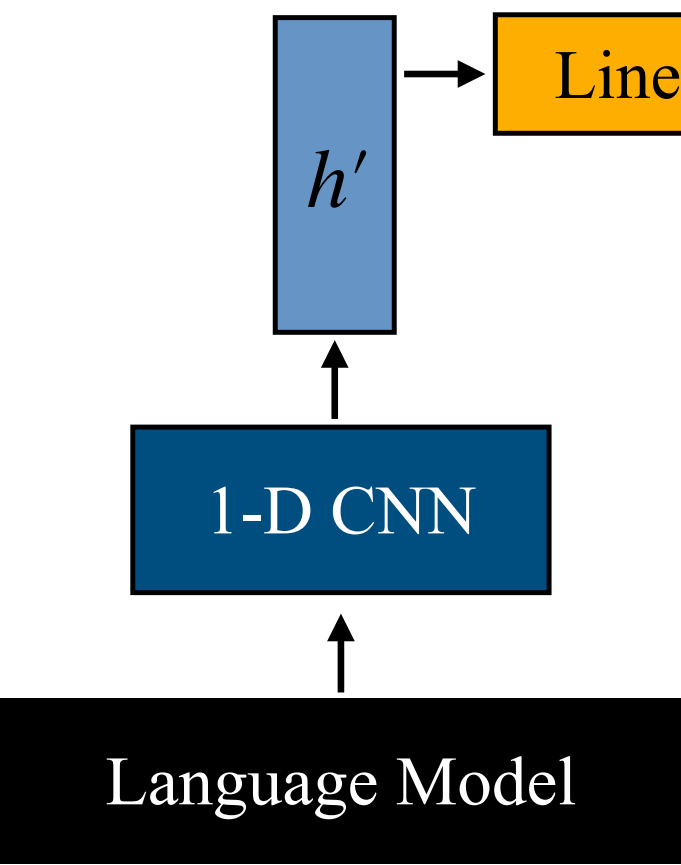
Train:
2nd match

Train:
3rd match

$q = 2$

Similarity, q .
Count of
consecutive
(depth-wise)
nearest matches
in training that are
correctly
predicted *and*
match $\hat{y}$ of the test
instance

Magnitude. Output
logits, $\mathbf{z}'$



Training: SDM LOSS

$$\mathcal{L} = -\frac{1}{N} \sum_n \log_{(2+q)} \left(\frac{(2+q)^{d \cdot z'_i}}{\sum_{c=1}^C (2+q)^{d \cdot z'_c}} \right)$$

Instance-wise change-of-base.

Models the transition from *epistemic* (reducible) uncertainty to *aleatoric* (irreducible) uncertainty

Alg. 1: Search Algorithm to Estimate a Region with Class- and Prediction-Conditional Accuracy at Least α

Seek to define a selection criteria that partitions the output distribution & that can be applied to new, unseen test points:

$$\text{SDM}_{\text{HR}} := \begin{cases} \hat{y} & \text{if } q' \geq q'_{\min} \wedge \text{sdm}(\mathbf{z}')_{\hat{y}} \geq \psi_{\hat{y}} \\ \perp & \text{otherwise} \end{cases}$$

SDM($\mathbf{z}'$)

$$q' = \min \left(q, (2 + q)^{\text{SDM}(\mathbf{z}')_{\hat{y}}} \right)$$

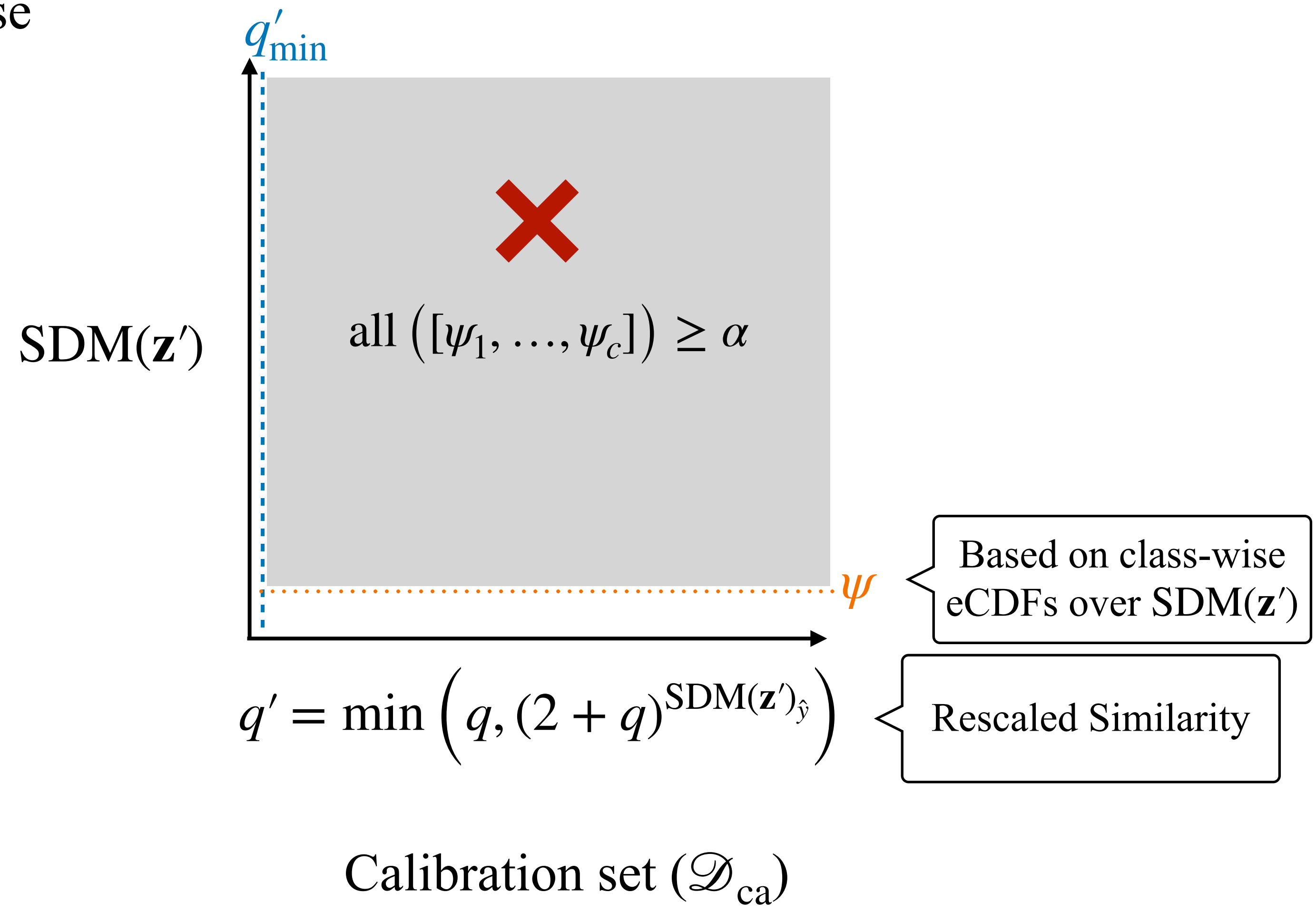
Rescaled Similarity

Calibration set ($\mathcal{D}_{\text{ca}}$)

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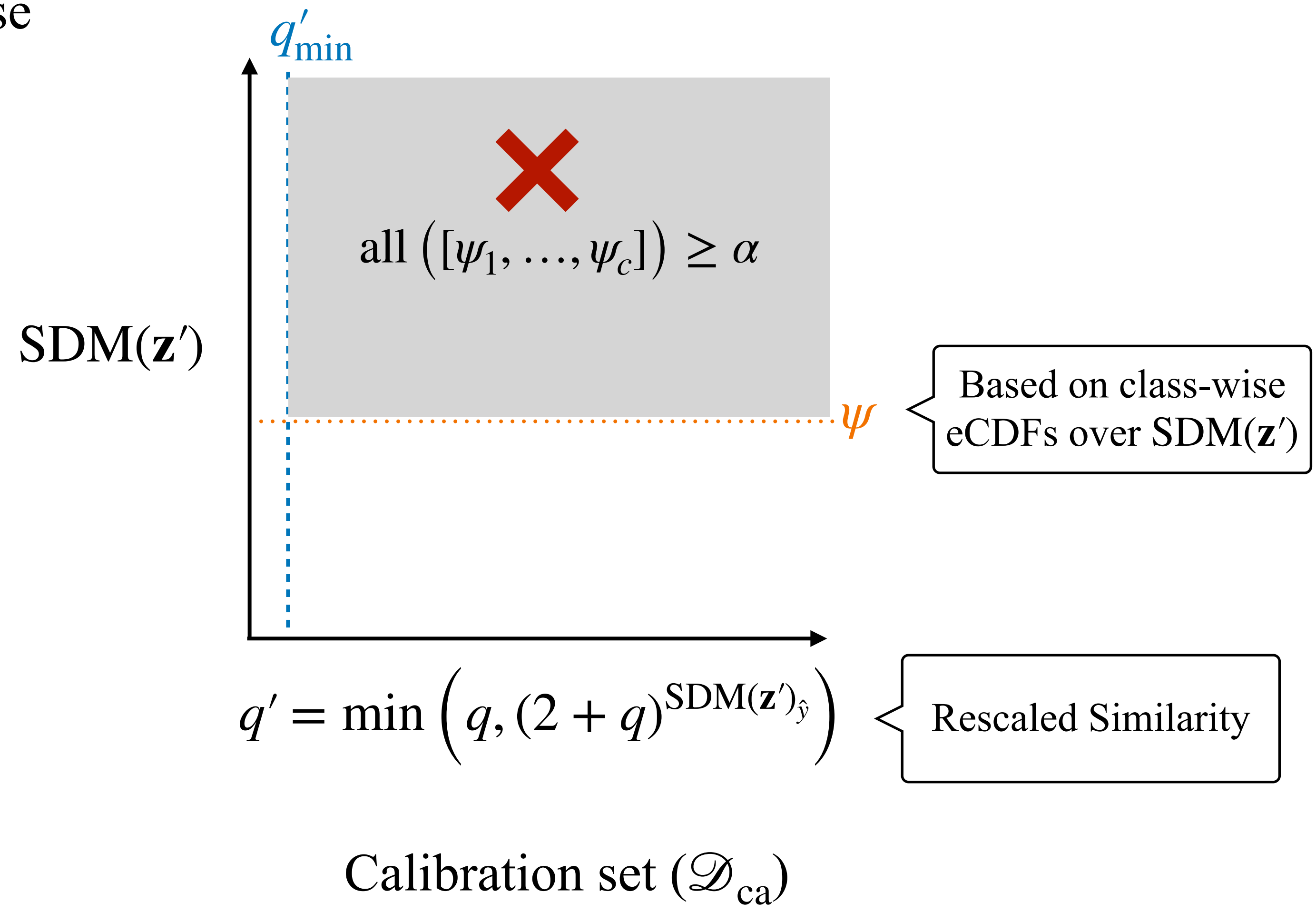
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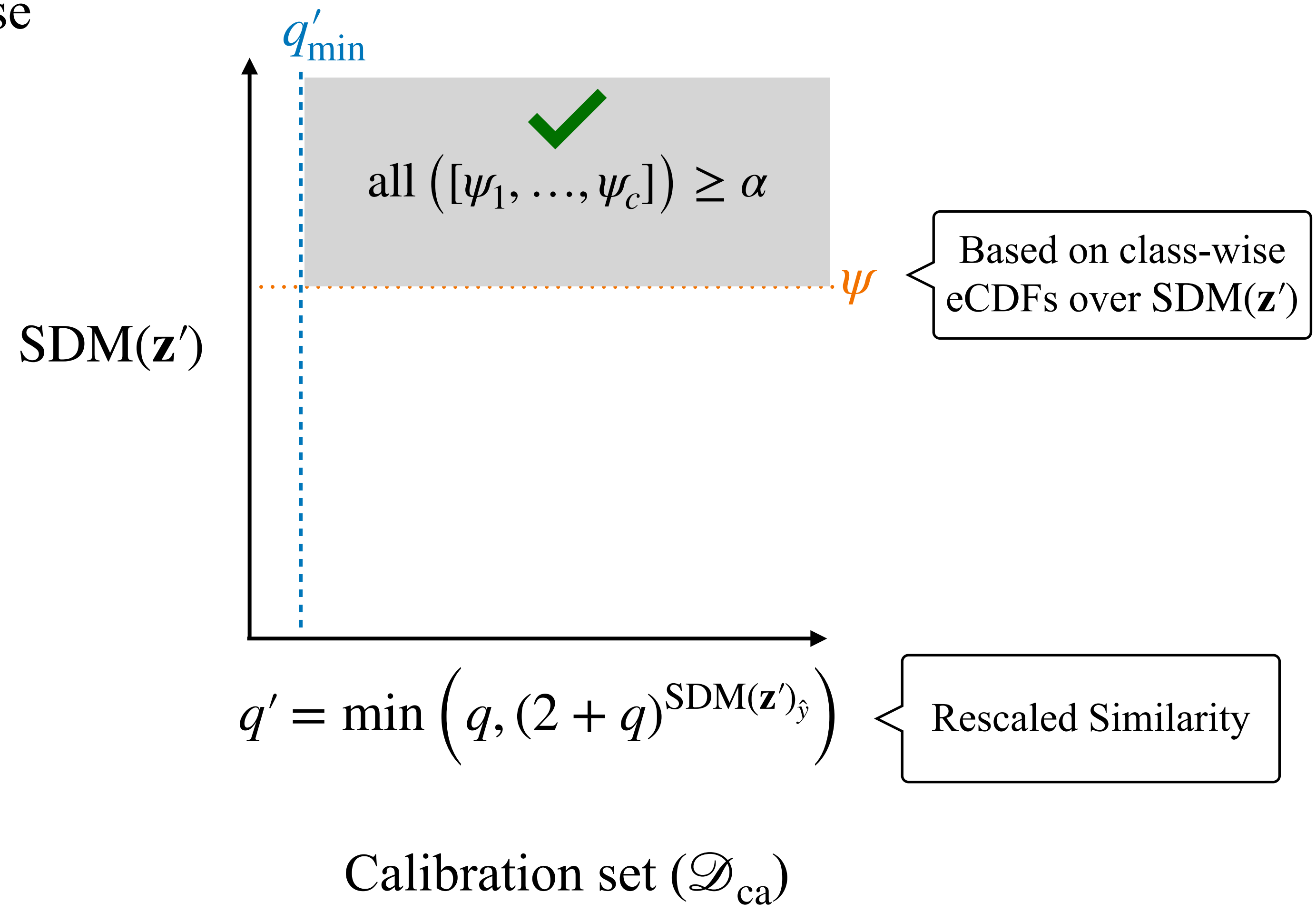
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We refer to the resulting region as the HIGH-RELIABILITY (SDM_{HR}) region.

SDM ESTIMATOR

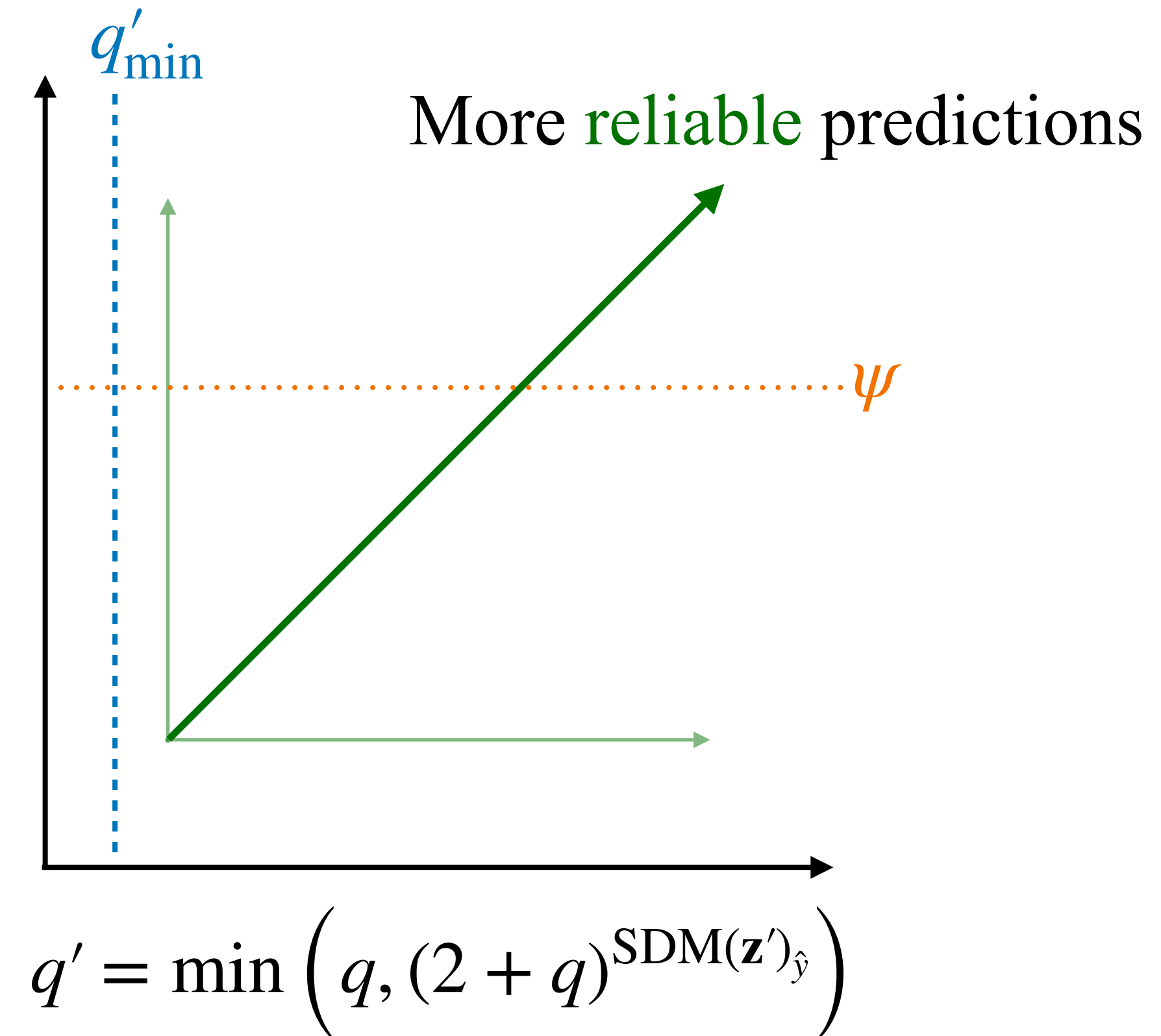
$$\text{SDM}_{\text{HR}} := \begin{cases} \hat{y} & \text{if } q' \geq q'_{\min} \wedge \text{sdm}(\mathbf{z}')_{\hat{y}} \geq \psi_{\hat{y}} \\ \perp & \text{otherwise} \end{cases}$$

SDM ESTIMATOR.

Data-driven partitioning of the calibration set to estimate a region with class- and prediction-conditional accuracy at least α , a value near 1

$\text{SDM}_{\text{HR}}^{\text{lower}}$ models the effective sample size (see Appendix A.5)

$\text{SDM}(\mathbf{z}')$



For some applications, it may be useful to calculate multiple α values, successively excluding each higher region when running Alg. 1. A point would then be assigned to the region of the most conservative (i.e., closer to 1) α in which it is a member.

Experiments

Task: Binary Sentiment Classification.

$y \in \{0 = \text{negative}, 1 = \text{positive}\}$

Eval sets:

SENTIMENT: In-distribution

SENTIMENTOOD: Out-of-distribution

Task: Binary Factuality Classification.

$y \in \{0 = \text{false}, 1 = \text{true}\}$

Eval set:

FACTCHECK: Covariate-shift

Calibration: Class- and prediction-conditional accuracy at least α , a value near 1 (here, $\alpha = 0.95$)

Model: Phi-3.5-mini-instruct

Estimator Comparisons: RAPS (conformal); VBLL (Bayesian); temperature scaling; and thresholding softmax (with and without adaptors)

Additional models, datasets, and comparisons in the paper

Results: In-distribution

Dataset	Model	Estimator	$y = 0$		$y = 1$		$\hat{y} = 0$		$\hat{y} = 1$		$y \in \{0,1\}$	
			Acc.	$\frac{n}{ \mathcal{D}_{te} }$	Acc.	$\frac{n}{ \mathcal{D}_{te} }$	Acc.	$\frac{n}{ \mathcal{D}_{te} }$	Acc.	$\frac{n}{ \mathcal{D}_{te} }$	Acc.	$\frac{n}{ \mathcal{D}_{te} }$
SENTIMENT	PHI3.5	SOFTMAX	0.98	0.50	0.86	0.48	0.88	0.56	0.98	0.42	0.93	0.98
SENTIMENT	PHI3.5+ADAPTOR	SOFTMAX	0.99	0.42	1.00	0.42	1.00	0.42	0.99	0.42	0.99	0.84
SENTIMENT	PHI3.5+ADAPTOR	TEMPSCALING	0.99	0.42	1.00	0.41	1.00	0.42	0.99	0.41	0.99	0.83
SENTIMENT	PHI3.5+ADAPTOR	RAPS	0.98	0.45	0.98	0.44	0.98	0.45	0.98	0.44	0.98	0.89
SENTIMENT	PHI3.5+DISCVBLLMLP	VBLL	0.99	0.42	1.00	0.40	1.00	0.42	0.99	0.40	0.99	0.82
SENTIMENT	PHI3.5+SDM	SDM _{HR}	1.00	0.37	0.99	0.30	0.99	0.38	1.00	0.30	0.99	0.68

THE SDM estimator remains informative over in-distribution data, despite the additional robustness

Evaluation.

Class- and prediction-conditional accuracy ≥ 0.95 , with a preference for full rejection (R) over under-coverage

Results: Covariate shifts and Out-of-distribution data



			$y = 0$		$y = 1$		$\hat{y} = 0$		$\hat{y} = 1$		$y \in \{0,1\}$	
Dataset	Model	Estimator	Acc.	$\frac{n}{ \mathcal{D}_{te} }$	Acc.	$\frac{n}{ \mathcal{D}_{te} }$	Acc.	$\frac{n}{ \mathcal{D}_{te} }$	Acc.	$\frac{n}{ \mathcal{D}_{te} }$	Acc.	$\frac{n}{ \mathcal{D}_{te} }$
SENTIMENTOOD	PHI3.5	SOFTMAX	1.00	0.50	0.54	0.46	0.70	0.71	0.99	0.25	0.78	0.96
SENTIMENTOOD	PHI3.5+ADAPTOR	SOFTMAX	0.57	0.03	0.96	0.07	0.84	0.02	0.85	0.07	0.85	0.09
SENTIMENTOOD	PHI3.5+ADAPTOR	TEMPSCALING	0.60	0.02	0.97	0.05	0.86	0.01	0.87	0.06	0.87	0.07
SENTIMENTOOD	PHI3.5+ADAPTOR	RAPS	0.48	0.13	0.82	0.18	0.66	0.10	0.68	0.22	0.68	0.32
SENTIMENTOOD	PHI3.5+DISCVBLLMLP	VBLL	0.89	0.03	0.96	0.04	0.93	0.02	0.94	0.05	0.94	0.07
SENTIMENTOOD	PHI3.5+SDM	SDM _{HR}	1	<0.01	1	<0.01	1	<0.01	1	<0.01	1	0.01



Dataset	Model	Estimator	$y = 0$		$y = 1$		$\hat{y} = 0$		$\hat{y} = 1$		$y \in \{0,1\}$	
			Acc.	$\frac{n}{ \mathcal{D}_{\text{te}} }$	Acc.	$\frac{n}{ \mathcal{D}_{\text{te}} }$	Acc.	$\frac{n}{ \mathcal{D}_{\text{te}} }$	Acc.	$\frac{n}{ \mathcal{D}_{\text{te}} }$	Acc.	$\frac{n}{ \mathcal{D}_{\text{te}} }$
FACTCHECK	PHI3.5	SOFTMAX	0.94	0.51	0.73	0.46	0.79	0.60	0.92	0.36	0.84	0.97
FACTCHECK	PHI3.5+ADAPTOR	SOFTMAX	0.40	0.08	0.99	0.33	0.89	0.04	0.87	0.37	0.87	0.41
FACTCHECK	PHI3.5+ADAPTOR	TEMPSCALING	0.38	0.07	0.99	0.29	0.86	0.03	0.88	0.33	0.88	0.36
FACTCHECK	PHI3.5+ADAPTOR	RAPS	0.36	0.18	0.98	0.35	0.89	0.07	0.74	0.46	0.76	0.53
FACTCHECK	PHI3.5+DISCVBLLMLP	VBLL	0.35	0.07	1	0.31	1	0.02	0.88	0.36	0.88	0.38
FACTCHECK	PHI3.5+SDM	SDM _{HR}	R	0	1	0.12	R	0	1	0.12	1	0.12

Evaluation.

Class- and prediction-conditional accuracy ≥ 0.95 , with a preference for full rejection (R) over under-coverage

Conclusion

A Memory Layer for Actionable Interpretability.

SIMILARITY-DISTANCE-MAGNITUDE (SDM) ESTIMATORS provide

- ① $\hat{p}(y \mid \mathbf{x})$, a robust estimate of the predictive uncertainty
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- ③ Local *updatability* via changes to $\mathcal{D}_{\text{tr}}$ and $\mathcal{D}_{\text{ca}}$.

Similarity-Distance-Magnitude Activations <https://arxiv.org/abs/2509.12760>